

Classifying and Predicting Degree Trajectories in Longitudinal Ego Networks

Omar Lizardo* David Hachen†

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Abstract

Personal ego networks undergo substantial restructuring during major life transitions, yet research often treats personal network size (degree) as a stable individual trait. Drawing on eight-wave panel data from the *NetHealth* Study tracking an undergraduate cohort from matriculation in August 2015 to senior graduation in May 2019 ($N = 457$), we use Latent Class Growth Analysis (LCGA) with repeated-measures Poisson finite mixture models to investigate whether individuals follow distinct degree evolution pathways across all eight semesters. Model selection identifies three distinct developmental trajectory classes: a stable cadre of “Network Conservers” (28.7%) who maintain large personal networks ($\bar{D} \approx 17$ –19 alters) throughout college; a modal group of “Moderate Winnowers” (39.8%) whose social circles contract steadily from 14.5 to 9.0 alters; and an “Accelerated Winnower” class (31.5%) that undergoes an aggressive early winnowing from 11.0 down to an intimate core of 4.8 alters. Multinomial logistic regression reveals that trajectory group membership is predicted primarily by baseline psychological dispositions (Extraversion, Generalized Trust, Openness) and ethnoracial status, rather than by parental socioeconomic status. Crucially, decomposing degree across four role relational layers shows that the aggregate contraction of collegiate networks represents the adaptive pruning of weak ties rather than an erosion of strong ties in the core personal network: daily activated ties (4.4–5.9 alters) and close, support-providing ties (10.5–12.0 alters) remain largely invariant across all four years.

Keywords: egocentric networks; degree trajectories; Big Five personality; latent class growth analysis; Poisson mixtures; tie strength; NetHealth study.

*Department of Sociology, University of California, Los Angeles. Email: olizardo@soc.ucla.edu.

†Department of Sociology, University of Notre Dame. Email: dhachen@nd.edu.

1 Introduction

Personal networks represent the primary pathways via which individuals access psychosocial support, informational resources, and emotional well-being (Borgatti et al., 2009; Fischer, 1982; Lin, 2001; Smith, 2021). Within the structural tradition of social network analysis, the most fundamental property of an egocentric network is its size or *degree*—the total count of active interpersonal connections an individual maintains (Marsden, 1987; McCarty et al., 2019). While classical cross-sectional studies documented substantial inter-individual variance in personal network size (Campbell and Lee, 1991; Marsden, 1987), theoretical accounts of network evolution have increasingly emphasized that personal degree is not a static, invariant parameter. Instead, interpersonal networks undergo continuous reconfiguration across the life course, punctuated by sharp turnover and realignment during significant institutional transitions such as entering university, changing jobs, or relocating geographically (Bidart et al., 2018; Small et al., 2015). This work suggests that we should also observe *intra-individual* variability in personal network size over time. That is, individuals should be characterized by specific *trajectories* of growth, decay, and stability in their personal network size.

Despite wide consensus that personal networks evolve dynamically, empirical inquiry into degree trajectories has faced persistent methodological and substantive challenges. On the one hand, evolutionary and cognitive models of human sociality—most notably Dunbar’s (1992; 1998) social brain hypothesis and recent formulations from complex systems researchers pointing to specific social signatures in the way individuals distribute attention across ties (Heydari et al., 2018; Saramäki et al., 2014)—posit that cognitive processing limits, emotional bandwidth, and temporal budgets impose strict upper bounds on personal network volume. Accordingly, individuals are expected to have characteristic relational capacities that remain stable over time, such that alter turnover does not modify the underlying topological distribution of interaction frequency or network scale. On the other hand, network analysts in sociology emphasize that relational opportunities are heavily structured by institutional foci, residential arrangements, and organizational contexts (Feld, 1982; Small, 2009; Wang et al., 2020). The transition into a residential university environment, for instance, initially inundates individuals with an abundance of potential ties, followed by subsequent sorting into specialized cliques, peer groups, and sub-communities (Sepulvado et al., 2020; Wang et al., 2020).

This study investigates the empirical implications of these competing perspectives by providing a full-panel investigation of degree trajectories across the entire undergraduate life course. Tracking an undergraduate cohort across all eight collegiate semesters from freshman matriculation in August 2015 through senior graduation in May 2019 ($N = 457$), we formulate

our analysis around three core questions. First, what is the overall empirical trajectory of personal network size across the complete collegiate life course, and does the widely observed contraction in total degree reflect an erosion of core supportive communities or an adaptive pruning of weak ties? Second, when modeled using principled count-data mixture likelihoods, what distinct latent trajectory classes emerge inductively across the undergraduate career? Third, what baseline sociodemographic characteristics and psychological dispositions predict which trajectory group a student will follow?

To answer these questions, we first decompose total ego network degree into four functional relational layers—total alter nominations, affective closeness, contact frequency, and functional social support—showing that while total degree contracts by roughly 24% over the college career, daily activated ties and close supportive circles remain exceptionally stable. Second, we fit Latent Class Growth Analysis (LCGA) with repeated-measures Poisson finite mixture models across the eight semesters, identifying three distinct developmental trajectory classes: *Network Conservers*, *Moderate Winnowers*, and *Accelerated Winnowers*. Third, we estimate multinomial logistic regression models to evaluate whether baseline sociodemographic background or psychological dispositions predict individual assignment to trajectory classes. Finally, in an Appendix, we demonstrate that continuous multilevel Poisson growth curve models corroborate our main findings, confirming that extraverts experience significantly steeper tie winnowing over time while generalized trust preserves overall network scale.

2 Theoretical Background and Hypotheses

2.1 Cognitive Constraints, Dunbar Layers, and Personal Network Dynamics

The structural scale of human sociality has long been theorized as bounded by biological and cognitive constraints. Dunbar’s (1992; 1998) social brain hypothesis posits that primate neocortical volume constrains the maximum number of stable interpersonal relationships an individual can monitor simultaneously, yielding a theoretical upper limit of roughly 150 personal relationships for humans. Within this outer boundary, egocentric networks exhibit a nested hierarchy of emotional closeness and contact frequency, typically organized into layers of approximately 5 support-clique intimates, 15 sympathy-group associates, 50 affinity-group contacts, and 150 total active acquaintances (Dunbar, 2018; Hill and Dunbar, 2003; Zhou et al., 2005).

In an extension of Dunbar’s framework, Saramäki et al. (2014) showed that individuals exhibit persistent *social signatures*—characteristic distributions of communication frequency

allocated across ranked ego-alter ties. Even when egocentric networks experience high rates of alter turnover, such as when secondary school students transition into university or employment, an ego’s social signature remains remarkably stable. Subsequent research has confirmed that these distributional signatures persist across multiple communication channels, including cellular voice calls and short-message services (Heydari et al., 2018).

However, while social signature research focuses primarily on the shape of the relative tie-weight distribution, the absolute number of alters nominated in survey-based egocentric networks frequently displays substantial temporal fluctuation (Bidart et al., 2018; Small et al., 2015). During periods of major institutional relocation, individuals enter new relational foci that supply novel alters while simultaneously weakening geographic proximity to prior contacts (Feld, 1982; Hachen et al., 2022; Lomi and Stadtfeld, 2014). This tension between cognitive-relational capacity and shifting institutional opportunities suggests that while some individuals may preserve a constant network size over time, others may undergo rapid expansion followed by subsequent contraction as weak ties are winnowed away.

2.2 Psychological Dispositions vs. Sociodemographic Stratification

When individuals navigate open social contexts, such as a residential university campus, what factors govern whether their personal networks expand, contract, or remain stable? Prior research points to two distinct classes of determinants: psychological pre-dispositions and sociodemographic background, the latter of which provides individuals with the relevant economic, relational, and cultural resources to facilitate network formation (DiMaggio and Garip, 2011).

2.2.1 Personality Factors

Psychological theories emphasize that personal network structure reflects broad individual differences in behavioral tendencies and interpersonal motivations, as captured by the Five Factor Model of personality (Costa and McCrae, 1992; Roberts et al., 2007). Extraversion, characterized by sociability, assertiveness, positive affect, and social reward-seeking behavior, has consistently been linked to larger overall network size, more frequent social interactions, and higher rates of tie initiation (Asendorpf and Wilpers, 1998; Centellegher et al., 2017; Harris and Vazire, 2016). Highly extraverted individuals have a stronger motivational orientation toward social engagement, which may encourage the accumulation of exploratory ties. Conversely, Neuroticism, reflecting emotional instability, negative affect, vulnerability to stress, and interpersonal rejection sensitivity, is frequently associated with smaller, more strained personal networks and higher rates of relationship dissolution (Russell et al., 2008).

Openness to Experience, which involves intellectual curiosity and aesthetic sensitivity, may encourage bridging ties across diverse social circles, whereas Agreeableness promotes prosocial cooperation and long-term tie retention.

2.2.2 Generalized Trust

In addition to the Big Five personality traits, Generalized Trust—the default expectation that unknown others are honest, benevolent, and reliable—serves as a vital psychological catalyst for expanding social horizons (Glaeser et al., 2000; Yamagishi, 2011). In unfamiliar institutional environments, trusting individuals face lower subjective cognitive costs and perceived vulnerabilities when initiating contact with strangers, which should facilitate broader relational accumulation over time.

2.2.3 Sociodemographic Factors

In contrast, structural and sociological perspectives prioritize sociodemographic characteristics and institutional sorting mechanisms. Stratification researchers have long shown that social network access and capital are structured by gender identity, racial/ethnic categorization, and family socioeconomic status (Campbell and Lee, 1991; Lin, 2001; Marsden, 1987). Historically, men and women have been observed to cultivate different egocentric network compositions, with women often maintaining networks with higher kin density and emotional support, and men reporting larger numbers of activity-based, non-kin ties (Moore, 1990). Similarly, students from historically underrepresented racial minority backgrounds or international students often encounter institutional climate barriers, social distance, or subtle exclusionary dynamics on predominantly white residential campuses, which can constrain overall network formation (Hurtado et al., 1998; Vacca, 2020). Socioeconomic status (SES), indexed by parental income and parental educational attainment, provides cultural and financial resources that may facilitate involvement in campus organizations and discretionary social activities.

2.3 Hypotheses

The theoretical traditions reviewed above offer contrasting expectations regarding how personal networks evolve across major life-course transitions and which individual factors govern developmental trajectories. On the one hand, structural stratification perspectives suggest that personal networks are fundamentally anchored in sociodemographic background, where parental economic capital and educational status equip students with resources that facilitate involvement in campus organizations and discretionary social life (Campbell and Lee, 1991; DiMaggio and Garip, 2011; Marsden, 1987). On the other hand, the institutional ecology of

a full-time residential university dramatically alters the relational playing field (Feld, 1982; Small, 2009). In a residential campus setting characterized by intense spatial co-presence, shared living arrangements, and equalized access to university infrastructure, formal status differences are partially backgrounded. Under these conditions, the formation, retention, and winnowing of personal ties should be driven primarily by endogenous psychological dispositions and behavioral orientations rather than inherited family background. Specifically, individual differences in interpersonal motivation (Extraversion), emotional reactivity (Neuroticism), and baseline social benevolence (Generalized Trust) directly regulate how students navigate relational opportunities and respond to cognitive and temporal bandwidth constraints. Synthesizing these arguments leads to our first formal expectation:

Hypothesis 1: *Baseline psychological dispositions (the Big Five traits and Generalized Trust) will be stronger predictors of degree trajectory group membership than baseline parental socioeconomic background.*

Among psychological dispositions, generalized trust plays a particularly critical role in governing network scale over time. In unfamiliar social contexts, individuals with low generalized trust experience heightened perceived vulnerability and steep subjective transaction costs when encountering strangers, fostering defensive relational withdrawal and rapid pruning of ties. Conversely, highly trusting individuals operate with a default expectation of benevolence, honesty, and reciprocity (Glaeser et al., 2000; Yamagishi, 2011). This disposition lowers the psychological friction required to initiate and maintain interpersonal relationships across diverse campus settings, encouraging sustained relational investment. Over the course of four collegiate years, this lower threshold for connection should protect against defensive winnowing, enabling trusting students to sustain large and durable personal networks:

Hypothesis 2: *Higher baseline Generalized Trust will be positively associated with membership in the stable Conserver trajectory class rather than the winnowing trajectory classes.*

In contrast, the evolutionary cognitive constraint perspective implies a dynamic tension between exploratory tie formation and long-term relational maintenance. Extraverted individuals are characterized by sociability, assertiveness, and heightened social reward-seeking, leading them to cast an exceptionally wide exploratory net upon entering college (Asendorpf and Wilpers, 1998; Costa and McCrae, 1992; Harris and Vazire, 2016). However, human cognitive processing limits and finite temporal budgets impose strict constraints on the number of relationships an individual can actively sustain (Dunbar, 1992; Saramäki et al., 2014). As collegiate life progresses and academic, organizational, and career demands escalate,

maintaining dozens of ties becomes cognitively and logistically prohibitive. Because extraverts accumulate the largest initial volume of casual connections, they face the greatest pressure to prune their networks back toward manageable cognitive boundaries. Consequently, while extraversion promotes early network expansion, it should paradoxically be associated with the steepest subsequent rate of tie winnowing:

Hypothesis 3: *Extraverted individuals will cast a wider initial exploratory net upon entering college, but will experience significantly steeper rates of subsequent tie winnowing over time compared to introverted peers.*

2.4 Tie Closeness, Functional Support, and the Weak-Tie Hypothesis

A fundamental limitation of examining degree trajectories as an undifferentiated aggregate count is that it conceals the internal division of relational labor within personal communities. In his foundational formulation of network structure, [Granovetter \(1973\)](#) distinguished between *strong ties*—characterized by a combination of emotional intensity, intimacy, mutual confiding, and reciprocal services—and *weak ties*, which involve superficial acquaintance, lower emotional investment, and infrequent or context-dependent interaction. As [Marsden and Campbell \(1984\)](#) demonstrated in their classic empirical assessment of tie strength, emotional closeness serves as the single best indicator of tie strength, whereas contact frequency is heavily conditioned by organizational foci and physical co-presence ([Feld, 1982](#)). Dormmates or classmates may interact daily simply due to spatial proximity rather than affective depth, whereas close friends or family may interact less frequently yet provide indispensable support ([Lizardo, 2024](#); [Smith, 2021](#)).

Furthermore, social support exchanges are highly specialized across distinct relational domains ([Wellman and Wortley, 1990](#)). While casual ties may provide companionship or shared recreation (“hanging out”), deeper forms of support—such as emotional comfort when distressed or personal advice during crises—are concentrated within a small, trusted core of strong ties. Drawing on Dunbar’s ([1992](#); [2018](#)) cognitive layering framework alongside Granovetter’s tie-strength perspective, we argue that the longitudinal contraction of collegiate networks does not represent an undifferentiated decay of social capital. Rather, we hypothesize that students actively protect their emotionally close, supportive, and daily activated strong ties, concentrating relational attrition almost exclusively among weak ties:

Hypothesis 4: *The longitudinal decline in collegiate network size will be concentrated primarily among weak ties, while strong ties (affectively close, daily activated, and support-providing ties) will remain invariant across the eight collegiate waves.*

3 Data and Analytical Sample

3.1 The NetHealth Project

Data for this study come from the *NetHealth* study (<https://sites.nd.edu/nethealth/>), a longitudinal study tracking an entire undergraduate cohort at the University of Notre Dame from matriculation in August 2015 through senior graduation in May 2019 (Liu et al., 2018; Sepulvado et al., 2020; Wang et al., 2020). The *NetHealth* study was designed to investigate the reciprocal co-evolution of social network structures, health behaviors (physical activity, sleep patterns, stress), and academic performance. Participants were recruited prior to entering college and were surveyed repeatedly across their undergraduate careers, completing comprehensive baseline surveys alongside semester-by-semester egocentric network batteries.

3.2 Longitudinal Analytical Cohort

Our analytical cohort is defined across all eight waves of the study (Wave 1 in Fall 2015 to Wave 8 in Spring 2019). The longitudinal cohort consists of $N = 457$ undergraduate students who completed at least three full egocentric network survey waves across the four-year undergraduate timeline, completed the initial baseline attributes survey, and maintained active participation. For fully adjusted multinomial predictive models, complete baseline covariate data are available for $N = 433$ participants.

3.3 Variables

3.3.1 Ego Network Degree and Relational Layers

In each survey administration, participants completed an egocentric network generator that asked them to nominate their most important social contacts (friends, family members, romantic partners, university peers), with an upper bound of 25 alters per wave. Total degree D_{it} for ego i at wave t is calculated as the total count of nominated alters ($D_{it} \in \{0, 1, \dots, 25\}$). Following name generation, respondents completed detailed tie-level evaluations assessing affective closeness, contact frequency, and functional social support. Affective closeness was reported on a four-point ordinal scale (“Especially Close,” “Merely Close,” “Less than Close,” or “Distant”), with alters designated as especially close or merely close defining the ego’s strong tie subnetwork. Contact frequency was similarly assessed across four categories (“Daily,” “Weekly,” “Monthly,” or “Less than Monthly”), with alters contacted on a daily basis constituting the daily activated tie layer.

Functional social support was evaluated using four binary indicators assessing whether

each nominated alter provided companionship (“hanging out”), academic or personal advice, emotional comfort when upset or distressed, or financial assistance. Alters providing at least one form of support constitute the ego’s overall supportive subnetwork. Crucially, the social support questions were not introduced into the NetHealth survey battery until Wave 2 (freshman spring) and were temporarily rotated out during Wave 6 (junior spring) due to survey modularization. As a consequence, supportive tie degree is available across Waves 2, 3, 4, 5, 7, and 8, but cannot be calculated for Wave 1 or Wave 6; hence, no data marker appears for support ties at Wave 1 in longitudinal trajectory figures.

3.3.2 Baseline Psychological Traits

All independent predictor variables were measured at baseline prior to or during Wave 1 matriculation in August 2015, establishing clear temporal ordering relative to subsequent network trajectories. Personality dispositions were measured using the 44-item Big Five Inventory (BFI) on five-point Likert scales, capturing Extraversion, Neuroticism, Agreeableness, Conscientiousness, and Openness to Experience ([John and Srivastava, 1999](#)). In all analytical models, each Big Five personality dimension is standardized as a z -score ($\mu = 0, \sigma = 1$) to facilitate direct comparison of effect sizes. Generalized Trust was assessed using a validated multi-item scale measuring default expectations of interpersonal benevolence and honesty ([Yamagishi, 2011](#)), similarly standardized to a z -score. In addition, we record ego birth order, categorized into firstborn or only child, middleborn, and youngest child.

3.3.3 Sociodemographic Characteristics

We include a comprehensive battery of sociodemographic covariates measured at baseline to account for structural sorting and social background across the undergraduate cohort. Gender identity is coded as a binary indicator: women (1) and men (0). Racial and ethnic background is categorized as White, Asian, Black, Hispanic/Latino, and Other or International. Family socioeconomic status is captured through three distinct indicators: parental annual household income, measured on an eight-point ordinal scale ranging from less than \$25,000 to \$250,000 or more, alongside maternal and paternal educational attainment, each measured on five-point ordinal scales ranging from less than high school to graduate or professional degrees. In addition, we account for international status and language background using binary indicators for U.S. citizenship (1 = citizen, 0 = non-citizen) and primary home language (1 = English, 0 = non-English). Finally, reflecting the institutional setting of a historically faith-based private residential university, religious affiliation is categorized into Catholic, Protestant, other religious tradition, and no religious affiliation (atheist, agnostic, or none).

4 The Collegiate Network Landscape

4.1 Aggregate Evolution and Functional Decomposition

To resolve the substantive meaning of personal network evolution across college, we first examine the empirical trajectory of ego network size across all eight collegiate waves, extending from matriculation in August 2015 through senior graduation in May 2019 ($N = 457$). Table 1 displays the empirical means and standard errors for Total Degree, Close Ties, Daily Activated Ties, and Support-Providing Ties across all eight semesters. Because the social support battery was first introduced into the NetHealth survey instrument during Wave 2 (freshman spring) and temporarily rotated out during Wave 6 (junior spring) due to survey modularization, support-providing degree cannot be computed for Waves 1 or 6; hence, these cells are designated with a dash in Table 1.

Table 1: Longitudinal Means and Standard Errors of Decomposed Relational Dimensions Across Eight Collegiate Waves ($N = 457$)

Academic Wave	Egos (N)	Total Degree (SE)	Close Ties (SE)	Daily Ties (SE)	Support Ties (SE)
Wave 1 (Frosh Fall)	323	14.19 (0.28)	13.00 (0.27)	5.93 (0.19)	—
Wave 2 (Frosh Spr)	442	13.38 (0.28)	12.46 (0.27)	6.93 (0.19)	11.99 (0.31)
Wave 3 (Soph Fall)	379	12.27 (0.33)	11.48 (0.31)	5.36 (0.20)	12.03 (0.32)
Wave 4 (Soph Spr)	371	11.37 (0.32)	10.82 (0.30)	5.92 (0.19)	11.08 (0.31)
Wave 5 (Jun Fall)	348	11.13 (0.33)	10.56 (0.32)	5.40 (0.18)	10.64 (0.32)
Wave 6 (Jun Spr)	300	10.64 (0.36)	10.14 (0.34)	4.93 (0.21)	—
Wave 7 (Sen Fall)	254	10.88 (0.39)	10.28 (0.38)	4.76 (0.21)	10.40 (0.38)
Wave 8 (Sen Spr)	181	10.80 (0.45)	10.04 (0.44)	4.38 (0.26)	10.45 (0.45)

Note: Standard errors in parentheses. Support questions were not administered in Wave 1 (freshman fall) or Wave 6 (junior spring); dashes indicate waves where the support battery was omitted.

Figure 1 visualizes the multi-line trajectory profiles of these four relational dimensions across all eight collegiate semesters.

The empirical pattern displayed in Figure 1 provides decisive support for **Hypothesis 4** and fundamentally clarifies the nature of collegiate network contraction. Total ego network size exhibits a steady downward trajectory, falling from 14.19 alters in freshman fall (Wave 1) to 10.64 in junior spring (Wave 6) and stabilizing at 10.80 in senior spring (Wave 8), representing an overall net contraction of approximately 24%. In sharp contrast, daily activated ties remain remarkably stable across all four years, peaking slightly in freshman spring (6.93 ties) before hovering consistently between 4.4 and 5.9 ties per wave across sophomore, junior, and senior years. Similarly, close ties and support-providing ties track

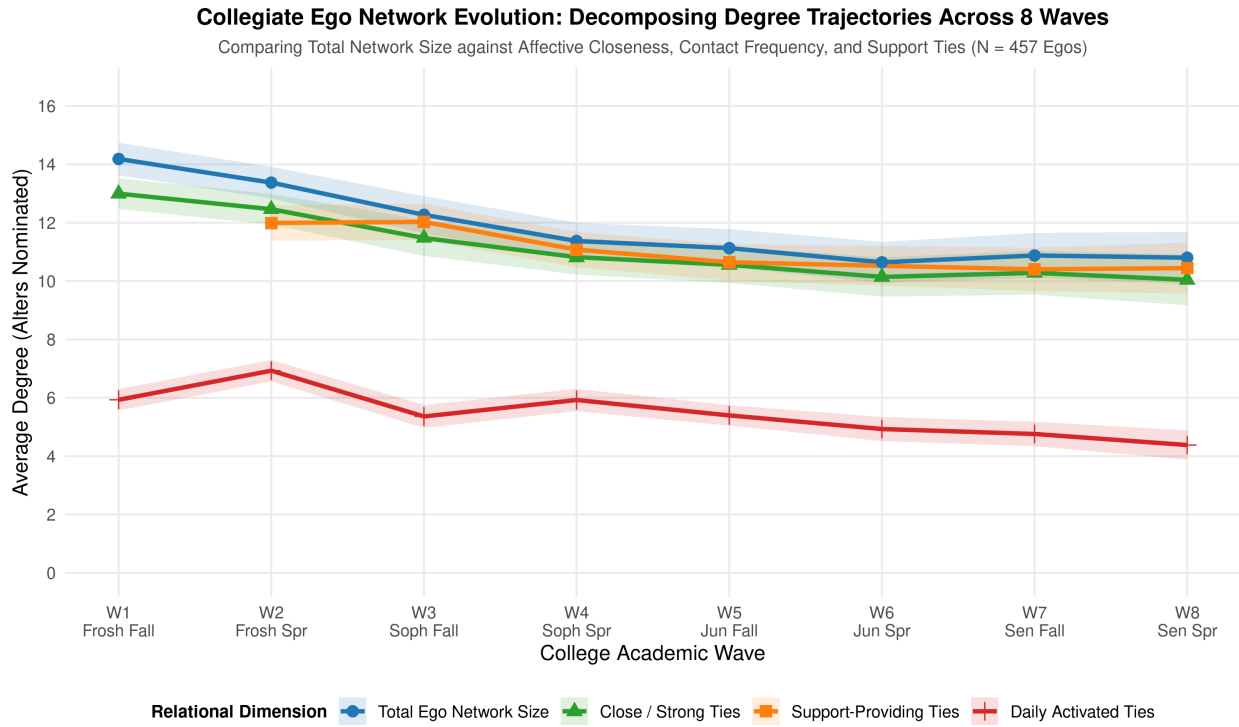


Figure 1: Decomposing ego network evolution across eight collegiate waves: Total Degree, Strong/Close Ties, Daily Activated Ties, and Support-Providing Ties ($N = 457$). Support questions were first administered in Wave 2 (freshman spring) and omitted in Wave 6; hence no data point appears for support ties at Wave 1. Shaded ribbons indicate 95% confidence intervals.

closely together, averaging approximately 10.5 to 12.0 ties across college.

Interpreted through Granovetter’s (1973) classic distinction between weak and strong ties, this functional decomposition reveals that the apparent downward degree trajectory widely documented in aggregate student surveys does not signify social isolation or an erosion of core supportive communities. Rather, students maintain a highly resilient, temporally stable inner core of strong ties, comprised of daily activated associates (matching Dunbar’s support clique of roughly 5 alters) and close, supportive ties (matching the sympathy group of roughly 10 to 12 alters). The aggregate decline in network size is driven almost entirely by the winnowing of weak ties that were temporarily accumulated during the initial disorganization of freshman year. Once students establish enduring campus niches, they prune weak ties to conserve cognitive and temporal bandwidth, while insulating their strong ties from decay.

4.2 Partitioning Social Support: Functional Specificity and Multiplexity

While the aggregate supportive subnetwork in Table 1 demonstrates general stability, treating any alter who provides at least one form of assistance as an undifferentiated supportive tie conceals meaningful variation in relational depth. Interpersonal support exchanges encompass qualitatively distinct functional domains, ranging from casual companionship to deep emotional vulnerability and material resource pooling (Marsden and Campbell, 1984; Wellman and Wortley, 1990). To examine whether personal networks can be partitioned further by support type and multiplexity, Table 2 decomposes the supportive subnetwork into four specific functional categories—companionship (hanging out), advice (academic and personal guidance), emotional comfort (comfort when upset), and financial assistance—alongside a structural partition by support multiplexity (uniplex ties providing exactly one support function versus multiplex ties providing two or more distinct supports).

The disaggregated support trajectories in Table 2 illuminate sharp empirical differences across functional domains. Companionship is the most pervasive support dimension, provided by 95.4% of all supportive ties. However, companionship ties decline steadily across college, falling from 11.29 alters in freshman spring to 9.48 alters by senior graduation, mirroring the broader contraction in discretionary social circles. In stark contrast, emotional comfort and advice support represent deeper forms of interpersonal investment that remain remarkably invariant over time. Emotional comfort—which requires mutual trust and affective vulnerability—hovers consistently between 6.14 and 6.63 alters across all surveyed waves (averaging 6.42 ties in Wave 2 and 6.50 ties in Wave 8), aligning with the classical boundary of Dunbar’s core sympathy group. Advice-providing ties similarly exhibit enduring stability,

Table 2: Longitudinal Means and Standard Errors of Specific Support Dimensions and Multiplexity Tiers Across Waves ($N = 457$)

Academic Wave	Egos	Hang Out	Advice	Comfort	Finance	Uniplex	Multiplex
Wave 2 (Frosh Spr)	313	11.29 (0.32)	7.18 (0.24)	6.42 (0.23)	1.63 (0.08)	4.17 (0.22)	7.82 (0.25)
Wave 3 (Soph Fall)	376	11.27 (0.32)	7.44 (0.25)	6.63 (0.24)	1.59 (0.07)	3.87 (0.22)	8.16 (0.26)
Wave 4 (Soph Spr)	369	10.33 (0.31)	6.81 (0.22)	6.47 (0.22)	1.77 (0.09)	3.53 (0.21)	7.55 (0.24)
Wave 5 (Jun Fall)	346	9.85 (0.32)	6.67 (0.24)	6.14 (0.23)	1.68 (0.07)	3.41 (0.21)	7.23 (0.25)
Wave 7 (Sen Fall)	250	9.63 (0.37)	6.89 (0.29)	6.14 (0.26)	1.80 (0.09)	2.92 (0.21)	7.49 (0.29)
Wave 8 (Sen Spr)	175	9.48 (0.44)	7.05 (0.35)	6.50 (0.33)	1.74 (0.09)	2.82 (0.25)	7.63 (0.36)

Note: Standard errors in parentheses. Support questions were not administered in Wave 1 (freshman fall) or Wave 6 (junior spring). Hang Out = Companionship; Comfort = Emotional comfort; Finance = Financial assistance. Uniplex ties provide exactly one support function (94.7% of which represent companionship alone); multiplex ties provide two or more distinct support functions.

hovering between 6.67 and 7.44 alters throughout college. Financial assistance, by contrast, is structurally restricted to a small, invariant core of 1.59 to 1.80 alters across all waves, primarily reflecting family members or romantic partners who provide material backing.

Crucially, partitioning ties by the *number* of support functions they provide reveals the structural mechanism underlying tie decay. Ties providing only a single support function (uniplex support ties) are overwhelmingly (94.7%) companionship ties that involve hanging out without deeper emotional confiding or advisory exchange. In Granovetter’s terminology, these uniplex connections function essentially as weak ties. Across the undergraduate career, uniplex support ties decline by 32.4%, falling from 4.17 alters in freshman spring to 2.82 alters at senior graduation. In sharp contrast, multiplex support ties ($S_{ij} \geq 2$)—which combine companionship with advice, emotional comfort, or material aid—constitute the student’s enduring strong-tie core. These multiplex ties exhibit zero net decay across the entire four-year observation window, averaging 7.82 alters in Wave 2, 8.16 in Wave 3, 7.55 in Wave 4, 7.23 in Wave 5, 7.49 in Wave 7, and 7.63 in Wave 8. Thus, relational winnowing during college operates with surgical specificity: students shed weak, uniplex activity ties while keeping their multiplex, strong supportive ties completely intact.

5 Latent Degree Trajectory Classes

5.1 Mixture Formulation and Model Selection

While aggregate trajectories reveal overall population trends, they mask substantial inter-individual heterogeneity in how students navigate relational winnowing. To identify distinct developmental pathways without imposing arbitrary boundary rules or Gaussian assumptions, we implement Latent Class Growth Analysis (LCGA) using repeated-measures Poisson finite mixture models (Grün and Leisch, 2008).

Specifying a Poisson emission distribution with linear and quadratic time terms centered at elapsed collegiate semester ($\text{Time}_{it} \in \{1, \dots, 8\}$), the conditional expectation of degree for ego i in latent class k is given by:

$$\log(\mathbb{E}[D_{it} \mid C_i = k]) = \beta_{0k} + \beta_{1k}\text{Time}_{it} + \beta_{2k}\text{Time}_{it}^2 \quad (1)$$

Observations are clustered at the ego level (|egoid), allowing the Expectation-Maximization (EM) algorithm to estimate class assignment probabilities while natively handling unbalanced panel observations across waves under missing-at-random assumptions. We evaluate candidate solutions from $K = 1$ to $K = 5$ latent classes.

Table 3 reports the information-theoretic fit statistics across candidate models.

Table 3: Latent Class Growth Analysis (LCGA) Model Fit Statistics Across Candidate Poisson Mixture Models on Eight-Wave Panel ($K = 1 \dots 5, N = 457$)

Latent Classes	Log-Likelihood	Par	AIC	BIC	ΔBIC
$K = 1$	-9627.1	3	19260.3	19277.9	4364.4
$K = 2$	-7863.6	7	15741.2	15782.3	868.8
$K = 3$	-7529.6	11	15081.1	15145.6	232.2
$K = 4$	-7434.6	15	14899.1	14987.1	73.6
$K = 5$	-7382.0	19	14802.0	14913.4	0.0

As shown in Table 3, information criteria improve dramatically when moving from a single population trajectory ($K = 1$) to multi-class mixtures, with BIC dropping by 4,132.3 points from $K = 1$ to $K = 3$. While BIC continues to decline through $K = 5$, the marginal gains diminish substantially beyond $K = 3$. To balance mathematical parsimony with substantive interpretability, we select the three-class solution ($K = 3$) as the optimal representation of collegiate trajectory morphology.

5.2 Latent Trajectory Profiles Across Eight Waves

Figure 2 displays the model-implied trajectories and observed ego sequences across all eight semesters for the optimal three-class solution.

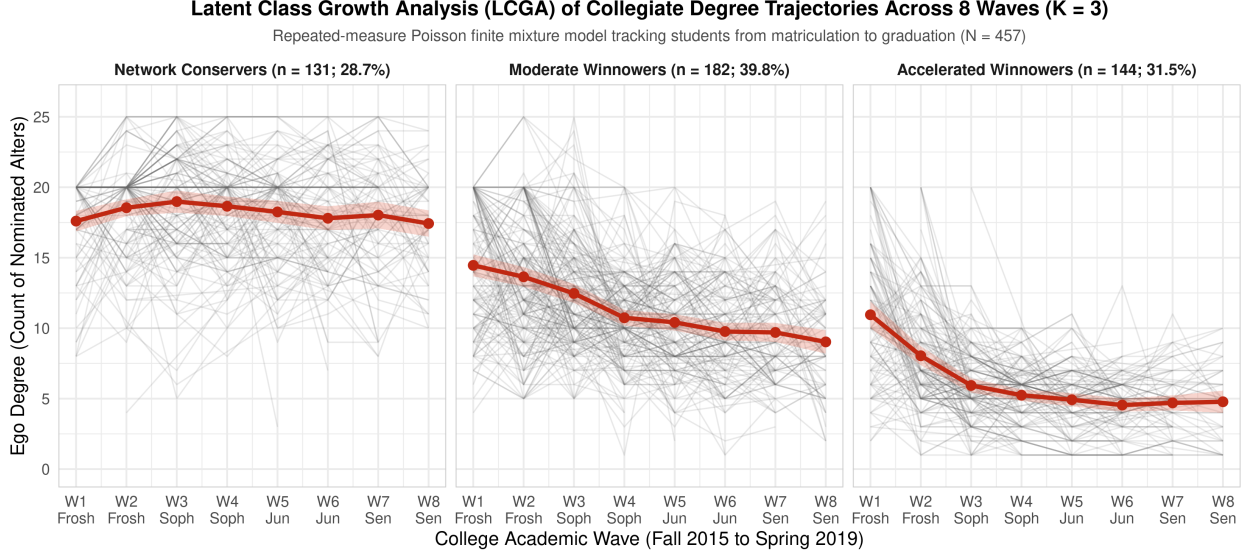


Figure 2: Latent Class Growth Analysis (LCGA) trajectory profiles from repeated-measures Poisson mixture model across eight collegiate waves ($K = 3, N = 457$). Thin grey lines denote individual ego trajectories; bold red lines and shaded ribbons indicate class mean trajectories and 95% confidence intervals.

The three latent classes capture distinct developmental archetypes of personal network evolution across the undergraduate career. First, **Network Conservers** ($n = 131, 28.7\%$) represent students who matriculate with large personal networks ($\bar{D}_1 = 17.60$) and preserve an exceptionally stable network scale throughout their entire undergraduate careers ($\bar{D}_8 = 17.43$), exhibiting essentially zero net winnowing over four years. Second, **Moderate Winnowers** ($n = 182, 39.8\%$) define the modal collegiate pathway, representing students who matriculate with average-sized circles ($\bar{D}_1 = 14.46$) and experience a steady, progressive winnowing across semesters, shedding roughly five to six alters by graduation ($\bar{D}_8 = 9.03$). Third, **Accelerated Winnowers** ($n = 144, 31.5\%$) enter college with somewhat smaller networks ($\bar{D}_1 = 10.95$), undergo an aggressive contraction across freshman and sophomore years, and stabilize at an intimate core of roughly 4 to 5 alters through senior graduation ($\bar{D}_8 = 4.78$).

5.3 Functional Tie Profiling Across Latent Classes

To ascertain whether winnowing classes experience an erosion of emotional closeness and social support, Figure 3 stratifies the functional tie decomposition (Total, Close, Daily, Support)

across the three latent LCGA classes across all eight waves.

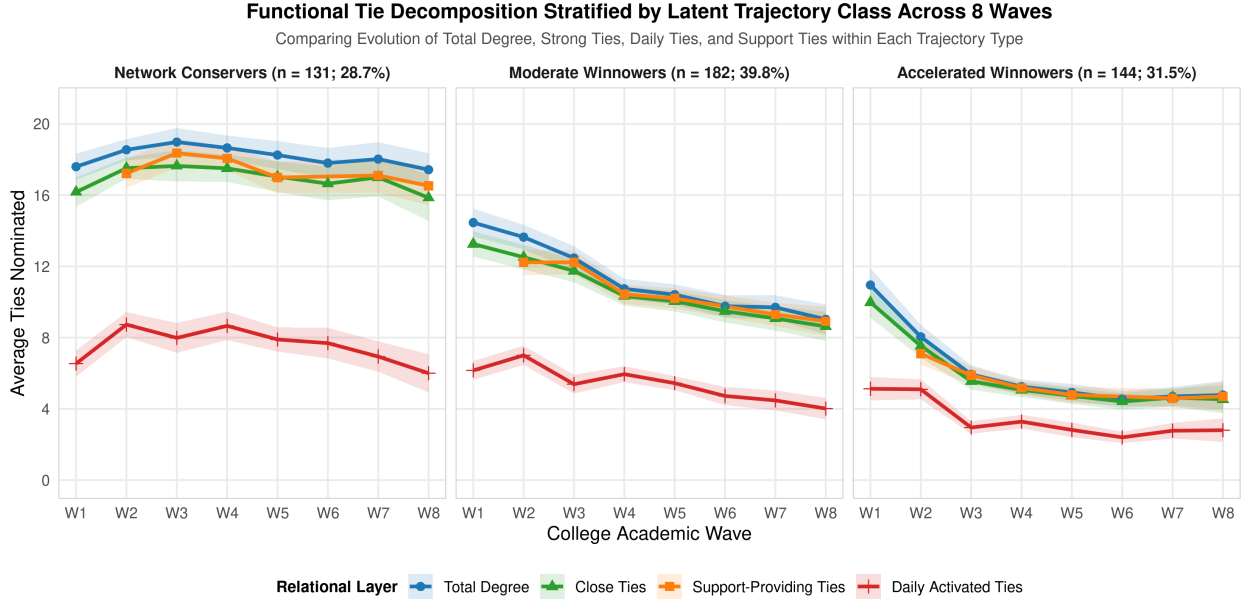


Figure 3: Functional tie decomposition (Total Degree, Close Ties, Daily Activated Ties, and Support-Providing Ties) stratified by latent trajectory class across all eight collegiate waves ($N = 457$). Shaded ribbons denote 95% confidence intervals.

Figure 3 provides compelling evidence regarding the internal mechanics of relational winnowing. For Network Conservers, high Total Degree (≈ 17 – 19 alters) is mirrored by high Close Ties (≈ 16 – 17.5) and high Support-Providing Ties (≈ 16.5 – 18.4), with daily activated ties remaining steady at roughly 7 to 9 alters. For Moderate Winnowers, Total Degree winnows from 14.5 down to 9.0 alters, but Close Ties (13.3 to 8.6) and Support Ties (12.2 to 8.9) track this contraction closely, while daily interaction stabilizes at roughly 4 alters. Most strikingly, for Accelerated Winnowers, the sharp contraction from 11.0 to 4.8 alters brings Total Degree, Close Ties, and Support Ties into perfect convergence around 4.5 to 5.0 alters by sophomore year. In other words, Accelerated Winnowers prune their weak ties until their total network coincides entirely with their strong, supportive core.

6 Predicting Trajectory Group Membership

Having established the three latent trajectory classes across all eight collegiate waves, we examine whether baseline sociodemographic characteristics and psychological dispositions predict which developmental trajectory an individual follows. We estimate multinomial logistic regression models using `nnet::multinom` on the complete-case cohort ($N = 433$), setting the modal category—**Moderate Winnowers**—as the reference category.

Table 4 reports the multinomial logit parameter estimates, standard errors, Odds Ratios (Relative Risk Ratios), 95% confidence intervals, and p -values for both Network Conservers and Accelerated Winnowers relative to Moderate Winnowers.

Figure 4 visualizes the Odds Ratios and 95% confidence intervals in a publication-grade forest plot.

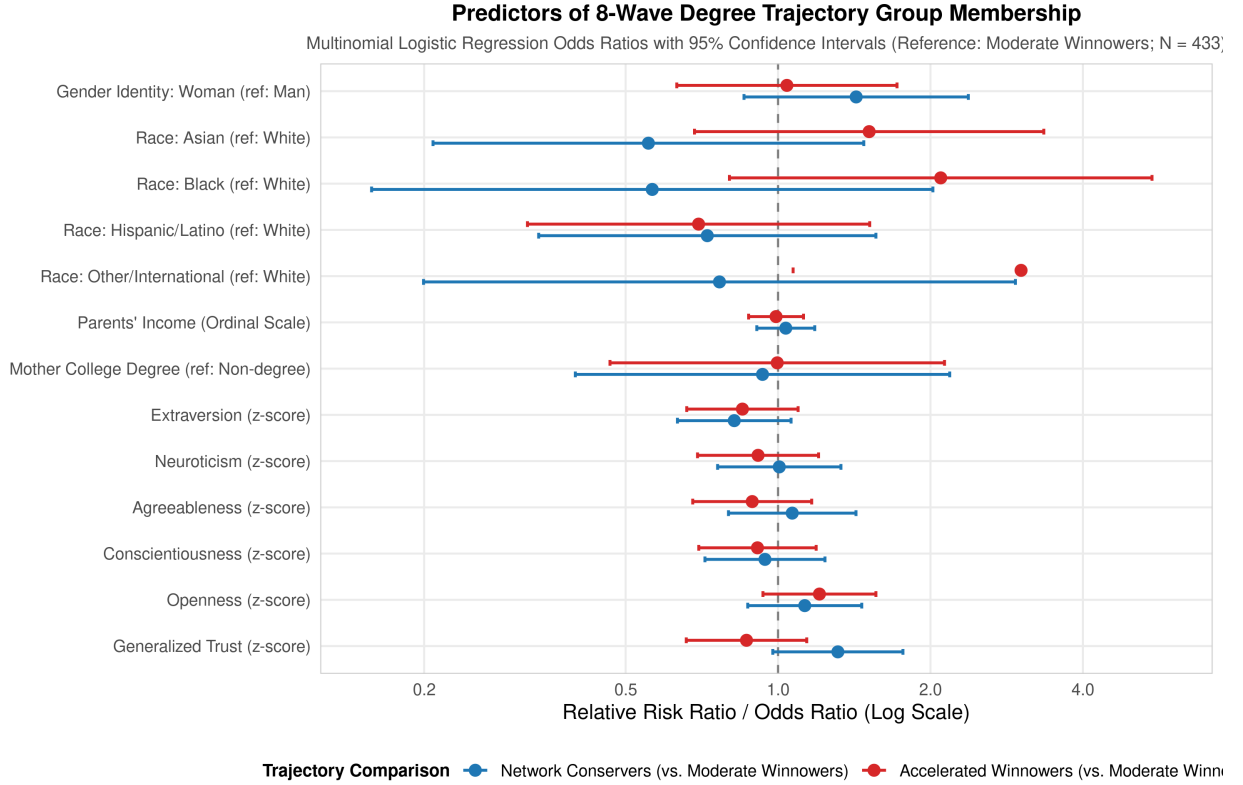


Figure 4: Forest plot of Odds Ratios (Relative Risk Ratios) from multivariable multinomial logistic regression predicting 8-wave latent trajectory class membership (Reference Category: Moderate Winnowers; $N = 433$). Error bars indicate 95% confidence intervals.

The multinomial regression results provide clear empirical evaluation of our core hypotheses. First, consistent with **Hypothesis 1**, classic indicators of family socioeconomic background—including parental income ($OR = 1.04, p = 0.601$ for Conservers; $OR = 0.99, p = 0.887$ for Winnowers) and maternal college degree ($OR = 0.93, p = 0.871$; $OR = 1.00, p = 0.993$)—exhibit zero predictive power. In contrast, psychological traits and trust display substantial substantive associations.

Second, supporting **Hypothesis 2**, higher baseline Generalized Trust increases the odds of belonging to the Network Conserver class by 31% per standard deviation ($OR = 1.31, 95\% CI: [0.98, 1.76], p = 0.072$), while reducing the likelihood of belonging to the Accelerated Winnower class ($OR = 0.87$). Trusting students are systematically more likely to

Table 4: Multinomial Logistic Regression Estimates Predicting Eight-Wave Latent Trajectory Class Membership (Reference Category: Moderate Winnowers; $N = 433$)

Predictor Variable	Estimate (SE)	Odds Ratio [95% CI]	<i>p</i> -value
<i>Panel A: Network Conservers (vs. Moderate Winnowers)</i>			
Gender Identity: Woman (ref: Man)	0.36 (0.26)	1.43 [0.86, 2.38]	0.172
Race: Asian (ref: White)	-0.59 (0.50)	0.55 [0.21, 1.48]	0.238
Race: Black (ref: White)	-0.57 (0.65)	0.56 [0.16, 2.02]	0.380
Race: Hispanic/Latino (ref: White)	-0.32 (0.39)	0.72 [0.34, 1.56]	0.411
Race: Other/International (ref: White)	-0.27 (0.69)	0.77 [0.20, 2.94]	0.698
Parents' Income (Ordinal Scale)	0.04 (0.07)	1.04 [0.91, 1.18]	0.601
Mother College Degree (ref: Non-degree)	-0.07 (0.43)	0.93 [0.40, 2.18]	0.871
Extraversion (<i>z</i> -score)	-0.20 (0.13)	0.82 [0.63, 1.06]	0.131
Neuroticism (<i>z</i> -score)	0.01 (0.14)	1.01 [0.76, 1.33]	0.969
Agreeableness (<i>z</i> -score)	0.06 (0.15)	1.07 [0.80, 1.43]	0.663
Conscientiousness (<i>z</i> -score)	-0.06 (0.14)	0.94 [0.72, 1.24]	0.669
Openness (<i>z</i> -score)	0.12 (0.13)	1.13 [0.87, 1.46]	0.358
Generalized Trust (<i>z</i> -score)	0.27 (0.15)	1.31 [0.98, 1.76]	0.072
<i>Panel B: Accelerated Winnowers (vs. Moderate Winnowers)</i>			
Gender Identity: Woman (ref: Man)	0.04 (0.26)	1.04 [0.63, 1.72]	0.875
Race: Asian (ref: White)	0.41 (0.41)	1.51 [0.68, 3.35]	0.307
Race: Black (ref: White)	0.74 (0.49)	2.10 [0.80, 5.48]	0.131
Race: Hispanic/Latino (ref: White)	-0.36 (0.40)	0.70 [0.32, 1.52]	0.363
Race: Other/International (ref: White)	1.11 (0.53)	3.02 [1.07, 8.51]	0.037
Parents' Income (Ordinal Scale)	-0.01 (0.06)	0.99 [0.87, 1.12]	0.887
Mother College Degree (ref: Non-degree)	-0.00 (0.39)	1.00 [0.47, 2.13]	0.993
Extraversion (<i>z</i> -score)	-0.16 (0.13)	0.85 [0.66, 1.10]	0.210
Neuroticism (<i>z</i> -score)	-0.09 (0.14)	0.91 [0.69, 1.20]	0.517
Agreeableness (<i>z</i> -score)	-0.12 (0.14)	0.89 [0.68, 1.17]	0.395
Conscientiousness (<i>z</i> -score)	-0.09 (0.14)	0.91 [0.70, 1.19]	0.492
Openness (<i>z</i> -score)	0.19 (0.13)	1.21 [0.93, 1.56]	0.151
Generalized Trust (<i>z</i> -score)	-0.14 (0.14)	0.87 [0.66, 1.14]	0.305

sustain large, durable networks across all four years.

Finally, with respect to racial background and institutional context, students identifying as Other or International face significantly higher odds of belonging to the Accelerated Winnower class ($OR = 3.02$, 95% CI: $[1.07, 8.51]$, $p = 0.037$), reflecting institutional friction and social distance on a predominantly white residential campus.

7 Discussion and Conclusion

7.1 Summary of Key Results

Taken together, the empirical findings across the eight-wave panel provide a unified, comprehensive portrait of how ego network size evolves across the undergraduate life course. Rather than supporting a singular narrative of continuous growth or uniform relational decay, the results reveal an adaptive social architecture characterized by pervasive, timing-dependent tie winnowing alongside remarkable temporal stability in core supportive communities.

First, evaluating functional relational layers across all eight semesters reveals that the widely observed aggregate degree contraction is an artifact of counting total nominations without differentiating tie strength. While total degree shrinks by 24% between freshman matriculation and senior graduation (falling from 14.19 to 10.80 alters), daily activated ties (4.4 to 5.9 alters) and close, support-providing ties (10.5 to 12.0 alters) remain invariant across all four years. Students actively protect their strong ties—their intimate support clique and sympathy group—while shedding weak ties. Furthermore, disaggregating the supportive subnetwork reveals that emotional comfort (hovering at 6.1 to 6.6 alters) and multiplex support ties (holding perfectly flat at 7.5 to 8.2 alters) remain completely impervious to decay, with net contraction concentrated exclusively in uniplex companionship ties.

Second, Latent Class Growth Analysis with Poisson mixtures establishes that students do not follow a uniform winnowing trajectory. Instead, undergraduate personal communities divide into three distinct developmental archetypes: a stable cadre of Network Conservers ($n = 131$, 28.7%) who preserve large personal networks (≈ 17.5 – 19.0 alters) throughout college; a modal group of Moderate Winnowers ($n = 182$, 39.8%) who matriculate with average-sized circles and steadily winnow down to ≈ 9 alters; and an Accelerated Winnower class ($n = 144$, 31.5%) that undergoes an early, rapid contraction to an intimate core of roughly 5 alters.

Third, multinomial logistic regressions confirm that trajectory group membership is governed primarily by baseline psychological dispositions and race rather than parental socioeconomic resources. Replicating the core insight of [Chandler and Hachen \(2018\)](#) within

a unified full-panel mixture framework, parental income and parental education exhibit zero predictive capacity, whereas Generalized Trust promotes retention in the stable Conserver class, and international/minority status accelerates winnowing. Finally, continuous growth modeling in the Appendix confirms that extraverts experience significantly faster rates of tie winnowing over time ($IRR = 0.994, p = 0.032$), showing that extraversion accelerates early exploratory tie accumulation rather than buffering against subsequent tie attrition.

7.2 Limitations and Suggestions for Future Work

A central methodological challenge in longitudinal personal network research is operationalizing ego network boundaries and tie strength across repeated panel waves. In the NetHealth study, network degree was measured via free-recall name generators bounded at a maximum of 25 alters, followed by name interpreter batteries assessing contact frequency, emotional closeness, and supportive exchanges. While bounding nominations at 25 alters captures the active sympathy group and minimizes respondent fatigue, it imposes an artificial ceiling that may truncate the personal networks of exceptionally expansive egos, potentially muting initial baseline differences in early college. Future investigations can address this constraint by using unbounded roster-based nomination instruments or hybrid designs that integrate self-reported ego networks with objective behavioral trace data, such as continuous Bluetooth proximity sensing, smartphone call and text logs, and campus Wi-Fi co-location metrics (Centellegher et al., 2017; Sepulvado et al., 2020).

Second, because our analyses rely on observational panel data from a single collegiate cohort, parameter estimates must be interpreted as conditional longitudinal associations rather than strict causal effects. Several potential threats to causal identification warrant consideration. Although baseline psychological dispositions and sociodemographic characteristics were measured prior to or during the initial network wave, unmeasured contextual confounding remains possible. For example, residential hall assignments, campus club participation, and academic major sorting represent institutional foci that structure local opportunity pools (Feld, 1982; Small, 2009). If extraverted or trusting students systematically select into high-density dormitories or collaborative extracurricular activities, part of the estimated trajectory slopes may reflect organizational context rather than psychological traits per se. Future studies should leverage natural experiments in student residential placement, paired with detailed administrative records on campus involvement, to disentangle dispositional tendencies from institutional sorting mechanisms.

Third, our modeling framework cannot fully eliminate the possibility of reciprocal feedback loops between network dynamics and psychological well-being over time. While we treat

personality traits as temporally antecedent, enduring individual dispositions (Costa and McCrae, 1992; John and Srivastava, 1999), personal network turnover can itself induce psychological stress, alter generalized trust, or reshape behavioral extraversion during formative life transitions (Asendorpf and Wilpers, 1998; Russell et al., 2008). Experiencing an acute contraction in social support during sophomore year may heighten neuroticism or erode generalized trust, generating a reinforcing spiral of relational withdrawal. Future research should implement continuous-time structural equation models, dynamic panel cross-lagged models, or latent curve models with structured residuals to trace the bidirectional co-evolution of personality traits and ego network structures across multiple years.

Fourth, several measurement granularity constraints of the survey design should be noted. Network surveys were administered once per semester (roughly two to three times per academic year). While this longitudinal rhythm effectively captures semester-to-semester macroscopic transitions, it cannot track the high-frequency micro-churn of tie activation, decay, and replacement that occurs across weeks or months within an academic term (Saramäki et al., 2014; Heydari et al., 2018). Furthermore, panel attrition across the four-year observation window reduced sample size in senior year (Wave 8, $n = 181$), reflecting survey fatigue and off-campus dispersal. Although our multilevel GLMMs and LCGA models naturally handle unbalanced panels under missing-at-random assumptions, future studies could employ mobile-app ecological momentary assessments (EMA) or monthly micro-surveys to capture high-resolution network churning while minimizing long-term attrition.

Finally, the empirical scope of this investigation is situated within a single, highly residential private university in the Midwestern United States characterized by a predominantly traditional-age, full-time undergraduate student body. The collegiate life course in this residential setting represents a unique institutional transition characterized by intense initial co-presence, communal living, and abrupt spatial reorganization upon graduation (Hurtado et al., 1998). Consequently, these findings cannot be assumed to generalize straightforwardly to commuter campuses, non-traditional student populations, working-class youth entering the labor market directly, or older adults navigating career changes or retirement. Future comparative research should extend this longitudinal trajectory framework to diverse institutional environments, working-adult cohorts, and cross-national contexts where personal community turnover is shaped by different organizational arrangements and life-course pacing (Bidart et al., 2018; Fischer, 1982).

7.3 Implications: The Adaptive Architecture of Personal Communities

Personal network analysis has long been animated by a foundational tension between structuralist social capital theories and evolutionary cognitive constraint models. In the classical social capital tradition (Granovetter, 1973; Lin, 2001; Borgatti et al., 2009), larger network degree is conceptualized as an unalloyed asset, providing access to diverse informational channels, instrumental sponsorship, and social support. From this perspective, the steady contraction of personal network size across college observed in empirical surveys might easily be interpreted as a problematic erosion of social integration or a deficit of relational connectivity. In contrast, the evolutionary cognitive constraint perspective pioneered by Dunbar (1992, 1998, 2018) posits that human sociality is biologically bounded by neocortical processing limits and finite temporal budgets, requiring individuals to distribute emotional and cognitive energy across nested hierarchical layers.

The empirical findings presented in this study challenge the pessimistic interpretation of network shrinkage by showing that longitudinal degree trajectories reflect an adaptive functional division of relational labor. Rather than signaling an undifferentiated decay of social ties, collegiate network contraction represents a deliberate, self-regulating process of relational winnowing. During the disorientation of freshman matriculation, students enter an unfamiliar social environment and temporarily cast a wide net, accumulating an expansive volume of weak ties. As academic demands intensify and enduring affinity groups coalesce, maintaining dozens of weak ties becomes cognitively and temporally unsustainable. Students resolve this constraint by winnowing weak ties while aggressively shielding their strong ties.

This functional insulation is clearly demonstrated by our eight-wave tie decomposition. Across all four undergraduate years, the volume of daily activated ties (averaging roughly 5 alters, precisely matching Dunbar’s support clique) and close, support-providing ties (averaging 10 to 12 alters, aligning with the sympathy group) remains exceptionally stable, even as total degree contracts by more than 25%. What appears in aggregate survey counts as network decay is actually the elimination of weak ties that contribute little to personal well-being or instrumental support. In this sense, collegiate network winnowing represents an adaptive optimization of personal communities, allowing individuals to concentrate finite social bandwidth where it yields the highest emotional and developmental returns.

Furthermore, our findings demonstrate that this winnowing process is fundamentally moderated by individual psychological dispositions. Personality traits serve as the underlying behavioral infrastructure that regulates how egos allocate relational labor and navigate institutional transition shocks. Extraverted students, driven by heightened reward sensitivity

and exploratory social tendencies, experience the most aggressive network winnowing, transitioning from an expansive initial circle to a lean, selective collegiate network. Generalized trust, by contrast, lowers perceived transaction costs and interpersonal friction, enabling trusting individuals to sustain larger overall networks across time without sacrificing core supportive stability (Yamagishi, 2011).

Ultimately, these dynamics suggest that longitudinal degree trajectories reflect the structural stabilization of personal communities into persistent social signatures (Saramäki et al., 2014; Heydari et al., 2018). Rather than passive recipients of structural churn, individuals act as purposeful architects of their social worlds, using dispositional tendencies to explore available opportunity structures and winnowing weak ties to preserve an enduring core of strong ties. By bridging formal latent growth mixtures, functional tie decomposition, and continuous multilevel modeling, this study provides a unified empirical foundation for understanding how personal networks adapt across major life transitions, opening new avenues for investigating the co-evolution of personality, social structure, and well-being across the adult life course.

Appendix: Continuous Multilevel Poisson Growth Curve Models (Robustness Check)

To verify that the discrete trajectory classes identified by LCGA reflect underlying continuous growth dynamics, we estimate multilevel Generalized Linear Mixed Models (GLMMs) with a Poisson response across all eight waves:

$$\log(\mathbb{E}[D_{it}]) = (\beta_0 + u_{0i}) + \beta_1 \text{Time}_{it} + \mathbf{X}_i \boldsymbol{\beta} + (\text{Time}_{it} \times \mathbf{Z}_i) \boldsymbol{\gamma} \quad (2)$$

where $u_{0i} \sim \mathcal{N}(0, \sigma_u^2)$ represents random ego intercepts capturing unobserved between-person heterogeneity, $\text{Time}_{it} \in \{0, 1, \dots, 7\}$ denotes elapsed semester centered at Wave 1 baseline, $\mathbf{X}_i$ is a vector of baseline sociodemographic and personality main effects, and $\mathbf{Z}_i$ captures cross-level interactions between focal personality traits (Extraversion and Neuroticism) and elapsed time.

Table A1 presents the model comparison fit hierarchy (Models 1–4) and the fully specified fixed effects estimates (Model 4).

Table A1: Fixed Effects Estimates from Multilevel Poisson Growth Curve GLMM with Random Ego Intercepts Across Eight Waves (Model 4; $N = 433$ egos, 2,456 observations)

Predictor Variable	Estimate	Std. Error	IRR [95% CI]	p-value
Intercept (Baseline Degree)	2.53	0.07	12.55 [10.90, 14.44]	< .001
Time (Elapsed Waves 0–7)	-0.06	0.00	0.95 [0.94, 0.95]	< .001
Extraversion (z -score)	0.02	0.02	1.02 [0.97, 1.07]	0.501
Neuroticism (z -score)	0.01	0.03	1.01 [0.96, 1.07]	0.609
Gender Identity: Woman (ref: Man)	0.01	0.04	1.01 [0.92, 1.10]	0.836
Race: Asian (ref: White)	-0.28	0.08	0.76 [0.65, 0.88]	< .001
Race: Black (ref: White)	-0.25	0.09	0.78 [0.65, 0.94]	0.008
Race: Hispanic/Latino (ref: White)	0.02	0.07	1.02 [0.89, 1.16]	0.773
Race: Other/International (ref: White)	-0.24	0.09	0.79 [0.66, 0.95]	0.011
Parents' Income (Ordinal Scale)	0.01	0.01	1.01 [0.99, 1.03]	0.325
Agreeableness (z -score)	0.04	0.02	1.04 [0.99, 1.09]	0.140
Conscientiousness (z -score)	0.00	0.02	1.00 [0.96, 1.05]	0.965
Openness (z -score)	-0.02	0.02	0.98 [0.94, 1.03]	0.442
Generalized Trust (z -score)	0.07	0.02	1.07 [1.02, 1.12]	0.006
Time $\times$ Extraversion	-0.01	0.00	0.99 [0.99, 1.00]	0.032
Time $\times$ Neuroticism	0.00	0.00	1.01 [1.00, 1.01]	0.101

Figure A1 visualizes the predicted degree trajectories across combinations of high versus low Extraversion (± 1 SD) and high versus low Neuroticism (± 1 SD).

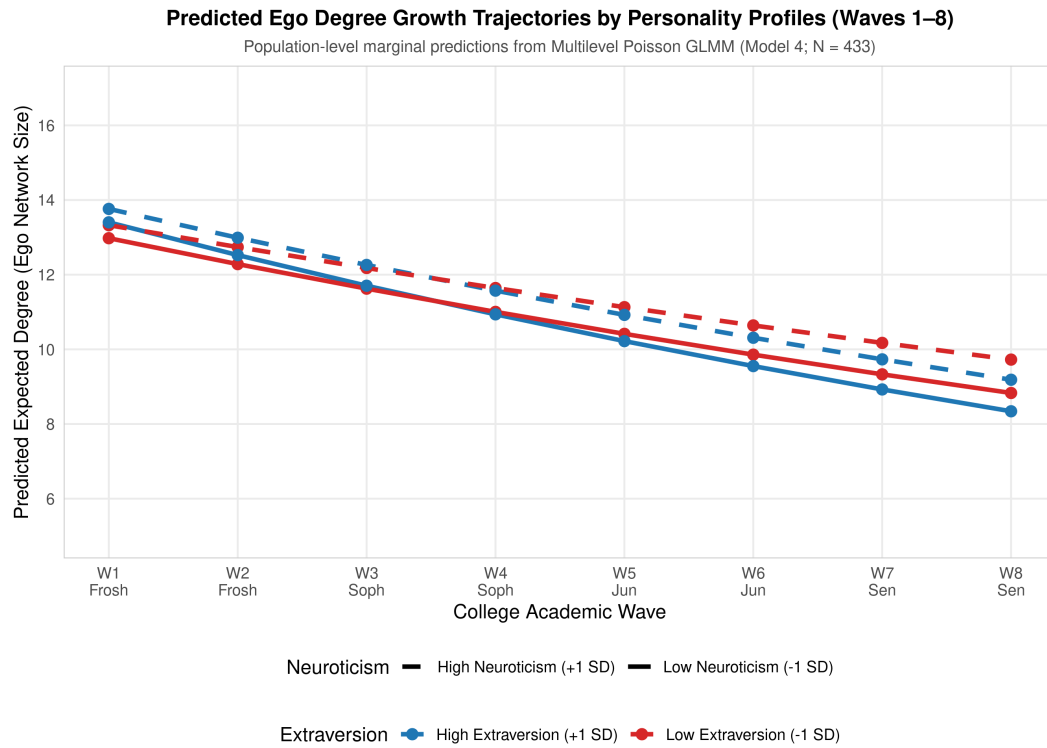


Figure A1: Predicted ego degree growth trajectories across eight collegiate waves by Extraversion (± 1 SD) and Neuroticism (± 1 SD) profiles estimated from Multilevel Poisson GLMM (Model 4; $N = 433$).

The continuous growth estimates directly corroborate our primary mixture findings. On average, student ego networks contract by 5.5% per elapsed collegiate wave (IRR = 0.945, $p < 0.001$). Baseline Generalized Trust exerts a significant positive main effect on network scale (IRR = 1.071, $p = 0.006$), confirming that trusting students sustain larger personal communities throughout college. Furthermore, the interaction between Time and Extraversion is statistically significant and negative (IRR = 0.994, $p = 0.032$). As shown in Figure A1, extraverted students enter college with larger initial networks (≈ 13.4 – 13.8 alters at Wave 1 vs. ≈ 12.9 – 13.3 for introverts) but undergo a faster rate of winnowing, crossing below their introverted peers by senior year (≈ 8.3 – 9.2 alters vs. ≈ 8.8 – 9.7 alters).

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